



Web Technologies

QP Code: SSC/Q8408

Version: 1.0

NSQF Level: 4.5

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Qualification Pack

Contents

SSC/Q8408: Web Technologies	3
<i>Brief Job Description</i>	3
Applicable National Occupational Standards (NOS)	3
<i>Compulsory NOS</i>	3
<i>Qualification Pack (QP) Parameters</i>	3
SSC/N8444: Work Organization and management for web technologies	5
SSC/N8445: Communication and interpersonal skills for web technologies	8
SSC/N8446: Design Implementation	11
SSC/N8447: Front-end Web development	14
SSC/N8448: Back End Development	17
Assessment Guidelines and Weightage	19
<i>Assessment Guidelines</i>	19
<i>Assessment Weightage</i>	20
Acronyms	21
Glossary	22

Qualification Pack

SSC/Q8408: Web Technologies

Brief Job Description

Individuals at this job are responsible for Web designing & development and maintaining web-based applications that include static and dynamic content. This includes the design, layout, and coding of a website. They may work independently or along with application/functional developers as part of the overall solution that includes a web-based component.

Personal Attributes

Individuals in this role must work and collaborate with various stakeholders involved in the development of web-based or mobile-based applications and solutions. They must be able to communicate and build relationships with others and continuously develop their knowledge and analytical abilities.

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SSC/N8444: Work Organization and management for web technologies](#)
2. [SSC/N8445: Communication and interpersonal skills for web technologies](#)
3. [SSC/N8446: Design Implementation](#)
4. [SSC/N8447: Front-end Web development](#)
5. [SSC/N8448: Back End Development](#)

Qualification Pack (QP) Parameters

Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Web & Mobile Development
Country	India
NSQF Level	4.5
Credits	17
Aligned to NCO/ISCO/ISIC Code	NCO-2015/ NIL

Qualification Pack

Minimum Educational Qualification & Experience	Completed 1st year of UG (UG Certificate) OR Pursuing 1st year of UG OR Previous relevant Qualification of NSQF Level (4) with 1-2 Years of experience relevant field
Minimum Level of Education for Training in School	Not Applicable
Pre-Requisite License or Training	NIL
Minimum Job Entry Age	19 Years
Last Reviewed On	NA
Next Review Date	08/02/2027
NSQC Approval Date	08/02/2024
Version	1.0
Reference code on NQR	QG-4.5-IT-01803-2024-V1-NASSCOM
NQR Version	1.0

Qualification Pack

SSC/N8444: Work Organization and management for web technologies

Description

Work Organization and management for web technologies

Scope

The scope covers the following :

- Work Organization and management for web technologies

Elements and Performance Criteria

Work organization and Management for web technologies

To be competent, the user/individual on the job must be able to:

- PC1.** Troubleshoot common web design and development problems
- PC2.** Work within specified time limitations and deadlines
- PC3.** Use a computer with a range of software packages
- PC4.** Apply research techniques and skills to keep up to date with industry best practices
- PC5.** Apply deployment optimization, such as page loading, with industry best practices
- PC6.** Ensure the work is completed according to a given schedule
- PC7.** Include linked images, fonts, native files, and production file format when archiving
- PC8.** Use software version control systems such as git

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Work organization and Management for web technologies</i>	30	70	-	-
PC1. Troubleshoot common web design and development problems	4	8	-	-
PC2. Work within specified time limitations and deadlines	4	8	-	-
PC3. Use a computer with a range of software packages	4	9	-	-
PC4. Apply research techniques and skills to keep up to date with industry best practices	4	9	-	-
PC5. Apply deployment optimization, such as page loading, with industry best practices	4	9	-	-
PC6. Ensure the work is completed according to a given schedule	4	9	-	-
PC7. Include linked images, fonts, native files, and production file format when archiving	3	9	-	-
PC8. Use software version control systems such as git	3	9	-	-
NOS Total	30	70	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8444
NOS Name	Work Organization and management for web technologies
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Web & Mobile Development
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2027
NSQC Clearance Date	08/02/2024

Qualification Pack

SSC/N8445: Communication and interpersonal skills for web technologies

Description

Communication and interpersonal skills for web technologies

Scope

The scope covers the following :

- Communication and interpersonal skills for web technologies

Elements and Performance Criteria

Communication and interpersonal skills for web technologies

To be competent, the user/individual on the job must be able to:

- PC1.** Read and understand specifications documents
- PC2.** Read and use provided source code of front-end and back-end technologies
- PC3.** Deliver products that respond to client requirements and specification
- PC4.** Gather, analyse, and evaluate information
- PC5.** Interpret standards and requirements
- PC6.** Match Client requirement
- PC7.** Present concepts to meet business requirements

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Communication and interpersonal skills for web technologies</i>	30	70	-	-
PC1. Read and understand specifications documents	4	10	-	-
PC2. Read and use provided source code of front-end and back-end technologies	4	10	-	-
PC3. Deliver products that respond to client requirements and specification	4	10	-	-
PC4. Gather, analyse, and evaluate information	4	10	-	-
PC5. Interpret standards and requirements	4	10	-	-
PC6. Match Client requirement	5	10	-	-
PC7. Present concepts to meet business requirements	5	10	-	-
NOS Total	30	70	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8445
NOS Name	Communication and interpersonal skills for web technologies
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Web & Mobile Development
NSQF Level	4.5
Credits	1
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2027
NSQC Clearance Date	08/02/2024

Qualification Pack

SSC/N8446: Design Implementation

Description

Design Implementation

Scope

The scope covers the following :

- Design Implementation

Elements and Performance Criteria

Design Implementation

To be competent, the user/individual on the job must be able to:

- PC1.** Create, analyse, and develop visual response to communication problems including understanding hierarchy, typography, aesthetics, and composition
- PC2.** Create, manipulate, and optimize images for the internet
- PC3.** Identify target markets and create concept for designs
- PC4.** Implement responsive designs that function correctly on multiple screen resolutions and/or devices.
- PC5.** Critique draft concepts, colour, and typography choices
- PC6.** Create wireframes, interactive prototypes, and design of user interfaces that consider user experience
- PC7.** Create code that conforms to and validates with the W3C standards including the accessibility guidelines
- PC8.** Create accessible and usable web interfaces for a variety of devices and screen resolutions
- PC9.** Use CSS or other external files to modify the appearance of web interfaces
- PC10.** Use CSS pre/post-processors
- PC11.** Create and update web interfaces for proper user experience and to assist with search engine performance
- PC12.** Use CSS to develop animations and interactive user interfaces

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Design Implementation</i>	30	70	-	-
PC1. Create, analyse, and develop visual response to communication problems including understanding hierarchy, typography, aesthetics, and composition	2	4	-	-
PC2. Create, manipulate, and optimize images for the internet	2	4	-	-
PC3. Identify target markets and create concept for designs	2	5	-	-
PC4. Implement responsive designs that function correctly on multiple screen resolutions and/or devices.	2	5	-	-
PC5. Critique draft concepts, colour, and typography choices	2	5	-	-
PC6. Create wireframes, interactive prototypes, and design of user interfaces that consider user experience	2	5	-	-
PC7. Create code that conforms to and validates with the W3C standards including the accessibility guidelines	3	7	-	-
PC8. Create accessible and usable web interfaces for a variety of devices and screen resolutions	3	7	-	-
PC9. Use CSS or other external files to modify the appearance of web interfaces	3	7	-	-
PC10. Use CSS pre/post-processors	3	7	-	-
PC11. Create and update web interfaces for proper user experience and to assist with search engine performance	3	7	-	-
PC12. Use CSS to develop animations and interactive user interfaces	3	7	-	-
NOS Total	30	70	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8446
NOS Name	Design Implementation
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Web & Mobile Development
NSQF Level	4.5
Credits	5
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2027
NSQC Clearance Date	08/02/2024

Qualification Pack

SSC/N8447: Front-end Web development

Description

Front-end Web development

Scope

The scope covers the following :

- Front-end Web development

Elements and Performance Criteria

Front-end Web development

To be competent, the user/individual on the job must be able to:

- PC1.** Create website animations and functionalities to assist in context explanations and add visual appeal
- PC2.** Create and update JavaScript code to enhance websites' functionality, usability, and aesthetics
- PC3.** Manipulate data and custom media with JavaScript
- PC4.** Create modular and reusable JavaScript code
- PC5.** Write documentation including comments in code
- PC6.** Use open-source JavaScript libraries
- PC7.** Manipulate graphical elements and content elements using JavaScript
- PC8.** Write test cases and execute automated testing for JavaScript implementation
- PC9.** Handle errors in code, debug code and fix bugs
- PC10.** Use the Linux CLI to interact with the remote server to deploy an application on the remote server
- PC11.** Document deployment steps for applications as required

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Front-end Web development</i>	30	70	-	-
PC1. Create website animations and functionalities to assist in context explanations and add visual appeal	2	5	-	-
PC2. Create and update JavaScript code to enhance websites' functionality, usability, and aesthetics	2	5	-	-
PC3. Manipulate data and custom media with JavaScript	2	5	-	-
PC4. Create modular and reusable JavaScript code	3	6	-	-
PC5. Write documentation including comments in code	3	7	-	-
PC6. Use open-source JavaScript libraries	3	7	-	-
PC7. Manipulate graphical elements and content elements using JavaScript	3	7	-	-
PC8. Write test cases and execute automated testing for JavaScript implementation	3	7	-	-
PC9. Handle errors in code, debug code and fix bugs	3	7	-	-
PC10. Use the Linux CLI to interact with the remote server to deploy an application on the remote server	3	7	-	-
PC11. Document deployment steps for applications as required	3	7	-	-
NOS Total	30	70	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8447
NOS Name	Front-end Web development
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Web & Mobile Development
NSQF Level	4.5
Credits	5
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2027
NSQC Clearance Date	08/02/2024

Qualification Pack

SSC/N8448: Back End Development

Description

Back End Development

Scope

The scope covers the following :

- Back End Development

Elements and Performance Criteria

Back End Development

To be competent, the user/individual on the job must be able to:

- PC1.** Manipulate data, making use of programming skills
- PC2.** Protect against security exploits
- PC3.** Integrate with existing code using APIs (Application Programming Interfaces), libraries & frameworks
- PC4.** Create or maintain database tables to support system requirements considering data normalization, key constraints and correct data types
- PC5.** Create code that are modular and reusable
- PC6.** Write documentation and comments in code
- PC7.** Write test cases and execute automated testing for back-end implementation
- PC8.** Handle errors in code, debug code and fix bugs
- PC9.** Use the Linux CLI to interact with the remote server to deploy an application on the remote server
- PC10.** Document deployment steps for applications as required
- PC11.** Configure the webserver for given applications

Qualification Pack

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Back End Development</i>	30	70	-	-
PC1. Manipulate data, making use of programming skills	2	5	-	-
PC2. Protect against security exploits	2	5	-	-
PC3. Integrate with existing code using APIs (Application Programming Interfaces), libraries & frameworks	2	5	-	-
PC4. Create or maintain database tables to support system requirements considering data normalization, key constraints and correct data types	3	6	-	-
PC5. Create code that are modular and reusable	3	7	-	-
PC6. Write documentation and comments in code	3	7	-	-
PC7. Write test cases and execute automated testing for back-end implementation	3	7	-	-
PC8. Handle errors in code, debug code and fix bugs	3	7	-	-
PC9. Use the Linux CLI to interact with the remote server to deploy an application on the remote server	3	7	-	-
PC10. Document deployment steps for applications as required	3	7	-	-
PC11. Configure the webserver for given applications	3	7	-	-
NOS Total	30	70	-	-

Qualification Pack

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8448
NOS Name	Back End Development
Sector	IT-ITeS
Sub-Sector	Future Skills
Occupation	Web & Mobile Development
NSQF Level	4.5
Credits	5
Version	1.0
Last Reviewed Date	08/02/2024
Next Review Date	08/02/2027
NSQC Clearance Date	08/02/2024

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

- Carry out assessments under realistic work pressures found in the normal industry workplace (or simulated workplace).
- Ensure that the range of materials, equipment, and tools that learners use are current and of the type routinely found in the normal industry workplace (or simulated workplace) environments.
- Assess that all tasks and functions are completed in a way, and to a timescale that is acceptable in the normal industry workplace.
- Assign workplace (or simulated workplace) responsibilities that enable learners to meet the requirements of the NOS.
- Productivity levels must be checked to ensure that it reflects those that are found in the work situation being replicated.
- Provide situations that allow learners to interact with the range of personnel and contractors found in the normal industry workplace (or simulated workplace).

Qualification Pack

Minimum Aggregate Passing % at QP Level : 70

(Please note: Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N8444.Work Organization and management for web technologies	30	70	0	0	100	5
SSC/N8445.Communication and interpersonal skills for web technologies	30	70	0	0	100	5
SSC/N8446.Design Implementation	30	70	0	0	100	25
SSC/N8447.Front-end Web development	30	70	0	0	100	25
SSC/N8448.Back End Development	30	70	0	0	100	40
Total	150	350	0	0	500	100

Qualification Pack

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training

Qualification Pack

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.

Qualification Pack

Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.
Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.